

Module 3 — DADA2 in Galaxy to phyloseq

BIOL 4810

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Goal: Process amplicon sequencing reads in Galaxy using DADA2, examine key outputs, then prepare the files needed for analysis in **phyloseq**.

Submission: Fill in answers, upload screenshots where prompted, then click **Prepare submission preview** and **Download as PDF (Print)**.

0) Your dataset

0A. Sample set / project name:

WCSMicrobiome

0B. Sequencing type:

16S V4 paired-end

0C. Metadata grouping variable you plan to analyze in phyloseq:

SampleSource

DADA2 workflow order

Use the Galaxy tools in this order:

```
plotQualityProfile → filterAndTrim → learnErrors → dada → mergePairs →  
makeSequenceTable → removeBimeraDenovo → assignTaxonomy → sequence  
counts
```

Reminder: The point of this workflow is to turn messy raw reads into a clean ASV table with taxonomy that can be analyzed ecologically in **phyloseq**.

Part 1 — Quality profiles

1A) Forward read quality profile

What is the purpose of this step?

The purpose of the forward read quality profile is to assess the quality of the reads. This allows for assistance when making decisions later on such as when choosing where to truncate the reads.

Upload screenshot: forward plotQualityProfile PDF ScreenShot...lityPlotProfile

Where does quality begin to noticeably decline in the forward reads?

Around 220bp.

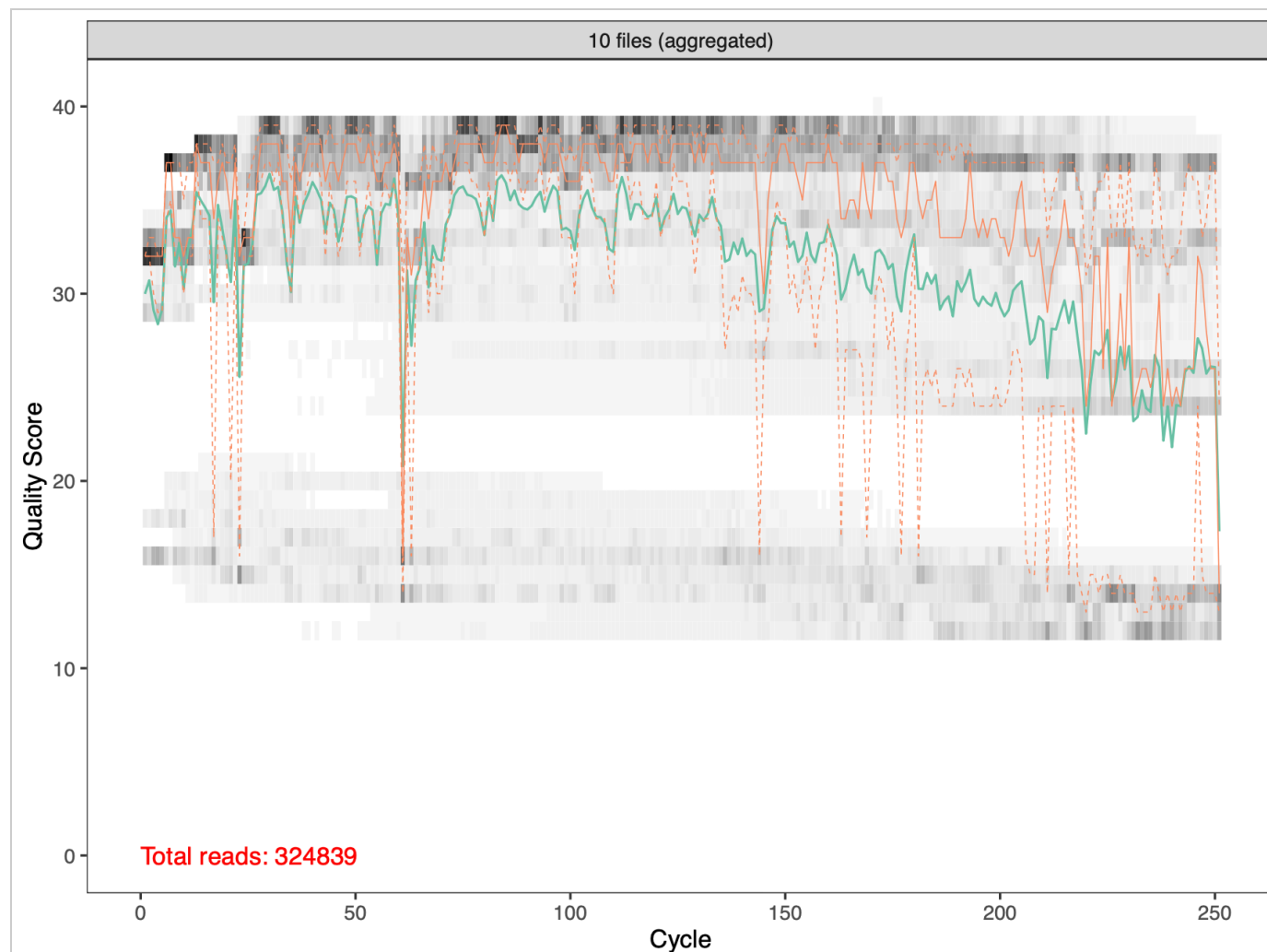
Chosen forward truncation length:

220bp

1B) Reverse read quality profile

What is usually different about reverse-read quality compared with forward-read quality?

Typically, the reverse is of poor quality as the reagents are not as fresh compared to the forward. This results in lower quality scores earlier in the reverse read compared to the forward.

Upload screenshot: reverse plotQualityProfile PDF
 ScreenShot...lityPlotProfile


Where does quality begin to noticeably decline in the reverse reads?

Around 205bp.

Chosen reverse truncation length:

205bp.

1C) Truncation decision

Explain why you chose these truncation values.

These truncation values were chosen because the areas of the plot that are of good quality have a quality score that is above 30. At the proposed values, the quality plots are consistently below a quality score of 30.

Rule of thumb: trim where quality begins to drop, but still keep enough overlap for forward and reverse reads to merge.

Part 2 — filterAndTrim

What filterAndTrim does:

This step removes low-quality reads and trims poor-quality bases from the ends of reads before denoising.

Why is this step important?

The step is important as it helps prevent low-quality reads from being considered in the downstream analysis. It ensures that the data is meaningful and accurate, preventing misleading conclusions regarding variation in the

Parameters used:

Truncate read length – 220; separate filters and trimmer for reverse reads – yes, trim start of each read – 10, truncate read length - 205.

What could happen if you trim too aggressively?

Trimming too aggressively can lead to shorter sequences and lost data (portions of the reads that are higher quality) leading to less overlap.

What could happen if you do not trim enough?

Trimming too little can result in more noise from the ends of the read that are of poor quality. For instance, portions of sequence from the adaptors at the ends would be included as well.

IMPORTANT — Split Paired Reads (Unzip Collection)

After `filterAndTrim`, your output is a **paired collection**.

Before continuing, you must split this into **forward and reverse collections**.

What to do in Galaxy

1. In the Tools panel, search: **unzip collection**
2. Input: your `filterAndTrim` paired output
3. Run

What you will get

- One collection = **forward reads (R1)**
- One collection = **reverse reads (R2)**

Why this matters

`learnErrors` and `dada` must be run **separately** on forward and reverse reads. If you skip this, `mergePairs` will fail.

Part 3 — learnErrors

What learnErrors does:

This step models the sequencing error rates in your dataset so DADA2 can distinguish real sequence variants

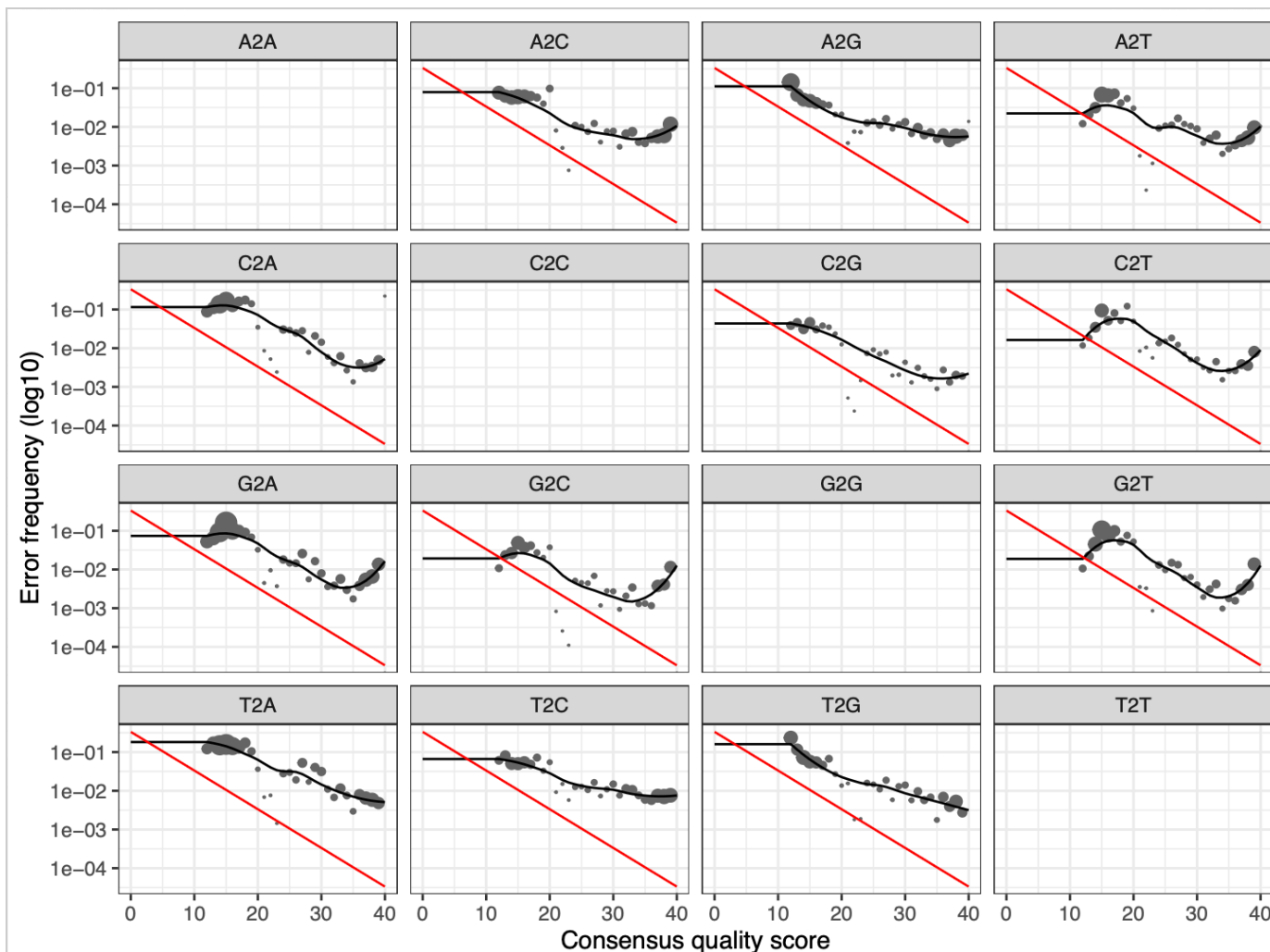
from sequencing mistakes.

Why is this step important?

This step is important as it enables dada2 to distinguish between sequences that are largely incorrect (or mistakes) and sequences that are of higher quality (genuine variants).

Upload screenshot: learnErrors PDF

Choose File ScreenShot...dLearnErrors



Does the fitted model generally follow the observed error pattern?

Somewhat.

Why is learning error rates important before denoising?

Learning error rates is important before denoising as the error rates reveal the rate at which bases are incorrect. Knowing this, any unnecessary noise from these errors can be minimized.

Part 4 — dada (denoising)

What dada does:

This is the denoising step. DADA2 uses the learned error model to correct likely sequencing errors and infer exact amplicon sequence variants (ASVs).

What is denoising doing?

Denoising is to fix errors in the sequence that were the result of PCR and sequencing. The output represents ASVs that were correct and should have errors that were resolved.

How is ASV inference different from OTU clustering?

OTU clustering is going to group together sequences that are similar to one another whereas ASV inference will correct and identify sequences.

Part 5 — mergePairs

What mergePairs does:

This step merges the forward and reverse reads for each amplicon so you can reconstruct the full sequence from overlapping read pairs.

What is being merged, and why?

The forward and reverse reads (fragments of a sequence) are being merged with the goal of forming a complete sequence.

Why do forward and reverse reads need to overlap?

The forward and reverse reads need to overlap because they are used to guide the reconstruction of the entire sequence using the matching portions of the sequence.

If reads fail to merge, what is one likely reason?

One likely reason could be in the trimming. Aggressive trimming could have removed too much of the reads, excluding the overlapping region necessary for the merging.

Part 6 — makeSequenceTable

What makeSequenceTable does:

This step creates the ASV table. Each row is a sample, each column is an ASV, and each cell contains the number of times that ASV was observed in that sample.

What does this table represent biologically?

This table represents the number of ASVs there are in the samples that were inputted. More specifically, it represents the different microorganisms present.

What are the rows, columns, and cell values in the sequence table?

The rows are the different organisms/samples, columns are the sequence variants, and the cell values are the number of instances each variant was found in that sample (row).

Part 7 — removeBimeraDenovo**What removeBimeraDenovo does:**

This step removes chimeras, which are artificial sequences created during PCR when fragments from different templates are incorrectly joined together.

What is a chimera, and why should it be removed?

A chimera are sequences that contain different fragments that should have not been paired together. They should be removed as they were errors and were not true variants.

Why are chimeras a problem for interpretation?

Chimeras are a problem for interpretation as they can unknowingly influence the conclusions made using the data. Since they contain fragments that should not have been joined together, they can lead to misleading

Part 8 — assignTaxonomy**What assignTaxonomy does:**

This step compares your ASVs to a reference database and assigns taxonomy, such as phylum, class, order, family, genus, or sometimes species.

What does taxonomy assignment add to your ASV table?

tells you the organisms that have the ASVs based on their database, allowing an individual to deduce what organisms may be present.

Why might some ASVs not be assigned all the way to species?

may not be entirely complete. Another reason is that multiple species may share a similar ASV (cannot definitively say an ASV matches one species more than another).

Part 9 — Sequence counts summary**What sequence counts shows:**

This output tracks how many reads remain after each major step in the pipeline so you can see where reads were retained or lost.

Why is it useful to track reads across steps?

Explain what this summary helps you evaluate.

Upload screenshot: sequence counts output

No file chosen

Approximately how many reads did you start with?

e.g., 45,000 per sample or similar

Approximately how many reads remained at the end?

e.g., 31,000 per sample or similar

At which step did you lose the most reads?

e.g., filtering, merging, chimera removal

What is one likely explanation for that loss?**Part 10 — Moving into phyloseq**

After completing DADA2 in Galaxy, we can create a phyloseq object for analysis.

10A. Which files will you take from DADA2 from Galaxy to use in phyloseq?

The files I will take from DADA2 from Galaxy to use in phyloseq include the ASV table (after using removeBimeraDenovo), the taxonomy table, and the metadata file.

10B. Why must sample names in the metadata match sample names in the ASV table?

relates the two sources of data. Without the matching names, the results of downstream analysis would be skewed.

10C. In 2–3 sentences, explain how DADA2 transforms raw reads into data that can be analyzed ecologically in phyloseq.

DADA2 transforms the raw reads into data that can be ecologically analyzed by first processing and merging the reads in addition to removing chimeras. This ensures that only portions of the sequences that are relevant are included in this process and aligned. DADA2 also then assigns taxonomy to the ASVs for interpretations and conclusions relating to the constituents of the microbiome.

Export (PDF)

1. Click **Prepare submission preview** (generates a clean summary).
2. Click **Download as PDF (Print)** and save.

Submission preview**Dataset**

Project/sample set: WCSMicrobiome

Sequencing type: 16S V4 paired-end

Grouping variable: SampleSource

Quality profiles

Purpose: The purpose of the forward read quality profile is to assess the quality of the reads. This allows for assistance when making decisions later on such as when choosing where to truncate the reads.

Reverse vs forward quality: Typically, the reverse is of poor quality as the reagents are not as fresh compared to the forward. This results in lower quality scores earlier in the reverse read compared to the forward.

Forward quality drop: Around 220bp.

Reverse quality drop: Around 205bp.

Forward truncation: 220bp

Reverse truncation: 205bp.

Why these values: These truncation values were chosen because the areas of the plot that are of good quality have a quality score that is above 30. At the proposed values, the quality plots are consistently below a quality score of 30.

filterAndTrim

Purpose: The step is important as it helps prevent low-quality reads from being considered in the downstream analysis. It ensures that the data is meaningful and accurate, preventing misleading conclusions regarding variation in the reads.

Parameters used:

Truncate read length – 220; separate filters and trimmer for reverse reads – yes, trim start of each read – 10, truncate read length – 205.

If trimmed too much: Trimming too aggressively can lead to shorter sequences and lost data (portions of the reads that are higher quality) leading to less overlap.

If not trimmed enough: Trimming too little can result in more noise from the ends of the read that are of poor quality. For instance, portions of sequence from the adaptors at the ends would be included as well.

learnErrors

Purpose: This step is important as it enables dada2 to distinguish between sequences that are largely incorrect (or mistakes) and sequences that are of higher quality (genuine variants).

Model fit: Somewhat.

Why important: Learning error rates is important before denoising as the error rates reveal the rate at which bases are incorrect. Knowing this, any unnecessary noise from these errors can be minimized.

dada + mergePairs

Denoising purpose: Denoising is to fix errors in the sequence that were the result of PCR and sequencing. The output represents ASVs that were correct and should have errors that were resolved.

ASVs vs OTUs: OTU clustering is going to group together sequences that are similar to one another whereas ASV interference will correct and identify sequences.

mergePairs purpose: The forward and reverse reads (fragments of a sequence) are being merged with the goal

of forming a complete sequence.

Need for overlap: The forward and reverse reads need to overlap because they are used to guide the reconstruction of the entire sequence using the matching portions of the sequence.

One reason reads fail to merge: One likely reason could be in the trimming. Aggressive trimming could have removed too much of the reads, excluding the overlapping region necessary for the merging.

Sequence table + chimera removal + taxonomy

Sequence table purpose: This table represents the number of ASVs there are in the samples that were inputted. More specifically, it represents the different microorganisms present.

Rows/columns/cells: The rows are the different organisms/samples, columns are the sequence variants, and the cell values are the number of instances each variant was found in that sample (row).

Chimera removal purpose: A chimera are sequences that contain different fragments that should have not been paired together. They should be removed as they were errors and were not true variants.

Why chimeras are a problem: Chimeras are a problem for interpretation as they can unknowingly influence the conclusions made using the data. Since they contain fragments that should not have been joined together, they can lead to misleading conclusions such as a believing there is an entirely sequence or organism.

Taxonomy purpose: Taxonomy assignment adds an identifier for the ASV relating to the varying levels of taxonomy. In other words, it tells you the organisms that have the ASVs based on their database, allowing an individual to deduce what organisms may be present.

Taxonomy limits: Some ASVs may not be assigned all the way to species because the reference database used for comparison may not be entirely complete. Another reason is that multiple species may share a similar ASV (cannot definitively say an ASV matches one species more than another).

Sequence counts

Purpose: -

Starting reads: -

Ending reads: -

Largest loss step: -

Likely explanation: -

Move to phyloseq

Files to download:

The files I will take from DADA2 from Galaxy to use in phyloseq include the ASV table (after using removeBimeraDenovo), the taxonomy table, and the metadata file.

Why names must match: Sample names in the metadata must match the sample names in the ASV table because this is how the software relates the two sources of data. Without the matching names, the results of downstream analysis would be skewed.

DADA2 to phyloseq transition: DADA2 transforms the raw reads into data that can be ecologically analyzed by first processing and merging the reads in addition to removing chimeras. This ensures that only portions of the sequences that are relevant are included in this process and aligned. DADA2 also then assigns taxonomy to the ASVs for interpretations and conclusions relating to the constituents of the microbiome.